

Aniket Dey

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Education

Dartmouth College

Hanover, NH

M.S. in Computer Science | GPA: 3.90/4.00

Sep 2027 - May 2028

- **Coursework:** Deep Learning, Graph-Based ML, DL Generalization & Robustness, Multimodal Generative AI

B.A. in Mathematics & Computer Science, Minor in Mathematical Finance | GPA: 3.70/4.00

Sep 2023 - May 2027

- **Coursework:** Statistical Data Science & Machine Learning, Algorithms, Software Design & Implementation, Operating Systems, Databases, Real Analysis, Probability, Linear Algebra, Differential Equations, Statistics

Experience

Citi Research

Jun 2026 – Aug 2026

Incoming Summer Analyst, Global Strategy & Macro Group

New York, NY

Newton Group, Dartmouth Department of Physics

Jan 2026 – Present

Undergraduate Research Assistant

Hanover, NH

- Engineered physical time-series features from multi-wavelength DECam light curves to detect rare transient flares in active galactic nuclei, improving same-object separation by 2.2x via grouped cross-validation against signal leakage

Dartmouth Department of Mathematics

Jan 2026 – Mar 2026

Teaching Assistant, MATH 86: Mathematical Finance I

Hanover, NH

- Supported 50+ students in weekly office hours & problem set reviews, reinforcing derivations of arbitrage-free pricing, Itô calculus, & Black-Scholes-Merton; drafted problem sets & graded exams evaluating pricing & hedging

Dartmouth Economics Department

Oct 2025 - Present

Undergraduate Research Assistant

Hanover, NH

- Engineered quantitative models in Python using Fama-French factor analysis to measure SEC regulatory effects in securities lending markets, calculating abnormal returns with 98% accuracy while isolating for short-sale constraints
- Applied time-series analysis using ARIMA & vector auto-regression to real-time options data to correlate returns tied to fails-to-deliver (FTDs) trades, examining 15,000+ trades for comprehensive market impact assessments

Amazon Web Services (AWS)

June 2025 – August 2025

Software Development Engineer Intern

Seattle, WA

- Built high-performance Python package to optimize query execution & latency on 100+TB data lakes, caching metadata as disk-level tree structures with logarithmic data retrieval & cutting query processing latency by 82%
- Designed scalable cache versioning framework with automated serialization protocols for distributed query management, reducing S3 compute costs by 54% while eliminating enterprise-scale cloud API bottlenecks

Hassanpour Lab, Dartmouth Center for AI & Precision Health

Sep 2024 – May 2025

Undergraduate Research Assistant

Hanover, NH

- Engineered distributed training architecture on Tensorflow leveraging CNNs & Vision Transformers for multi-modal pancreatic cancer diagnosis, achieving 90% clinical accuracy with real-time feature extraction from CT/MRI images
- Optimized inference pipelines using tensor graph optimization & custom batching; benchmarked latency-accuracy tradeoffs & stress-tested system performance across heterogeneous CT/MRI inputs to maintain sub-50ms latency

Oracle

Jun 2024 – Aug 2024

Software Engineer Intern

Redwood Shores, CA

- Architected database encryption protocol in SQL using hybrid RSA-ECC key management framework with modular arithmetic optimizations, reducing encryption overhead by 40% & achieving sub-millisecond digital key exchanges
- Built & deployed full-stack encryption tool to internal teams using Angular with OracleDB integration through custom JDBC API routes, utilizing multi-tiered query pipelines to decrease application response times by 35%

LISP Machine Learning Lab

Oct 2023 – May 2024

Undergraduate Research Assistant

Hanover, NH

- Developed adversarial linear programming algorithm employing SVD & convex optimization solvers to reverse engineer CTC-loss gradient, reconstructing 98% of original audio samples from training of large-scale ASR models
- Enhanced attack transferability across leading architectures through ensemble gradient aggregation techniques, achieving 85% cross-model success against Transformer & CNN models while maintaining reconstruction fidelity

Projects

ML Framework for Detecting Market Downturns From Implied Volatility | PyTorch, NumPy, MLFlow | Source Code [↗](#)

- Developed ensemble regression framework to predict market downturns from volatility skew analysis, achieving 91% AUC using time-series decomposition & regime-switching algorithms with vectorized Numpy operations
- Built high-performance ETL pipeline to process 1M+ options chain data points with sub-50ms latency, utilizing delta-neutral feature extraction to isolate pure skew in noisy market data for efficient, real-time risk detection

Heston Model Calibration Algorithm For SPY/SPX Surfaces | Python, SciPy, Pandas, Matplotlib | Source Code [↗](#)

- Engineered high-performance stochastic volatility calibration system, achieving 99% R^2 accuracy while reducing compute time by 65% through an optimized L-BFGS-B parameter search with parallel volatility surface fittings
- Implemented novel moment matching algorithm for initial parameters & real-time market data processing pipeline with sub-10ms model recalibration latency, outperforming standard FFT-only methods by 40% on live options flow

Options Pricing & Implied Volatility Surfaces Web Platform | Python, TypeScript, Pandas, Yfinance | Demo [↗](#)

- Built real-time pricing infrastructure integrating YFinance APIs with calibrated BS, Binomial & Monte Carlo models, enabling dynamic volatility fittings & price sensitivity analysis to inform hedging & portfolio management workflows
- Engineered historical backtesting engine with numerical differentiation algorithms for 1000+ concurrent contracts

Mamba-ESL: Optimizing SSM Sequence Modeling Architecture | Python, PyTorch, NumPy, CUDA | Source Code [↗](#)

- Designed architecture extending Mamba sequence models with embedding search & injection module to amplify relevant context linearly, achieving 35% inference speedup over Transformer baselines on long-context benchmarks
- Trained model with accelerated computing in CUDA using custom GPU kernels for attention masking & parallelism across hundreds of heads, integrating mixed-precision training & kernel fusion to optimize GPU runtime by 25%

Physics-Aware Diffusion World Model for Interactive Billiards | Python, PyTorch, CUDA, AWS | Source Code [↗](#)

- Built end-to-end diffusion transformer world model (1.52B params, 30 layers) trained on 400K synthetic billiards episodes with DDP on 2xH100 GPUs, implementing diffusion forcing to reduce long-horizon rollout error by 43×
- Engineered full ML pipeline with convolutional VAE for 48× latent compression, latent-space DiT with action conditioning & DDIM sampling, model distillation for low-latency inference, & real-time play via WebSocket

Martian: Agentic Infrastructure Optimization Startup | LangGraph, MCP, PyTorch, Node.js | Demo & Landing Page [↗](#)

- Co-founded & developed autonomous cloud optimization platform using LangGraph & MCP to orchestrate LLM agent integrations into major providers to eliminate idle infrastructure, saving \$15k/month across pilot deployments
- Developed custom workload prediction tools for agents using LSTM models for time-series forecasting for resource demand, utilizing optimized inference engines to enable real-time predictive scaling & achieve 23% cost reduction

Nuggets: Real-Time Multiplayer Game Engine | C, Linux, Bash, Valgrind | CS50 Software Design & Implementation Final

- Built online multiplayer mining game in C supporting 26 concurrent players with custom UDP network protocol, achieving sub-5ms message latency between clients through efficient socket programming & state synchronization
- Engineered line-of-sight visibility algorithm using ray-casting computations & dynamic 2D matrix updates, optimizing memory management with hash-table coordinate indexing to deliver <10ms real-time game rendering performance

Skills

Languages: Python, SQL, C++, Java

Technologies: PyTorch, TensorFlow, Pandas, Numpy, SciPy, React, Linux, PowerBI, Bloomberg Terminal, Excel

Skills: Quantitative Modeling, Machine Learning, Data Visualization & Analysis, Time-Series Forecasting, Databases